

The empirical recurrence for OEIS A264509: recovering a conjecture that is not written down

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Abstract

OEIS A264509 records “Empirical recurrence of order 69”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A264509 is “Number of $(3+1)X(n+1)$ arrays of permutations of $0..n*4+3$ with each element having directed index change -1,1 0,-1 0,1 or 1,0.”. It has offset 1 and begins

3, 4, 46, 211, 1235, 6613, 35356, 190821, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Nov 15 2015 18:40:45, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	5	c_{19}	−46090847	c_{37}	114923191	c_{55}	1487873
c_2	36	c_{20}	36730185	c_{38}	250443639	c_{56}	−3923298
c_3	−176	c_{21}	69059380	c_{39}	−83938401	c_{57}	−81220
c_4	−572	c_{22}	−47592894	c_{40}	−234007356	c_{58}	1031624
c_5	2881	c_{23}	−94616664	c_{41}	48616074	c_{59}	−67333
c_6	5073	c_{24}	35474627	c_{42}	183002698	c_{60}	−203151
c_7	−29109	c_{25}	130906196	c_{43}	−2629842	c_{61}	26399
c_8	−26399	c_{26}	2629842	c_{44}	−130906196	c_{62}	29109
c_9	203151	c_{27}	−183002698	c_{45}	−35474627	c_{63}	−5073
c_{10}	67333	c_{28}	−48616074	c_{46}	94616664	c_{64}	−2881
c_{11}	−1031624	c_{29}	234007356	c_{47}	47592894	c_{65}	572
c_{12}	81220	c_{30}	83938401	c_{48}	−69059380	c_{66}	176
c_{13}	3923298	c_{31}	−250443639	c_{49}	−36730185	c_{67}	−36
c_{14}	−1487873	c_{32}	−114923191	c_{50}	46090847	c_{68}	−5
c_{15}	−11390628	c_{33}	217898108	c_{51}	19178520	c_{69}	1
c_{16}	6830109	c_{34}	161146609	c_{52}	−25686170		
c_{17}	25686170	c_{35}	−161146609	c_{53}	−6830109		
c_{18}	−19178520	c_{36}	−217898108	c_{54}	11390628		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

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